

Project Proposal: ROI- and Semantics-Aware Feature Pruning for Brain-Aligned Visual Encoding

Abstract: This project investigates how visual representations are extracted from pretrained computer vision models aligning with human visual cortex responses measured with fMRI. Building on recent work such as VISGate and BOLDREAMS, we study whether compact and interpretable brain-aligned encoders can be obtained by selecting feature subsets that are specific to cortical regions of interest (ROIs) and to the semantic content of images. The project compares modern visual backbones such as DINOv2 and CLIP, predicts ROI-level fMRI responses on natural image datasets, and evaluates whether ROI-aware and semantics-aware pruning improves the trade-off between compactness, interpretability and predictive accuracy. The core pipeline focuses on ROI-wise encoding and feature pruning, while optional extensions include subject-aware analyses and lightweight gating strategies.

Dataset: Natural Scenes Dataset (NSD) or Algonauts-style visual cortex data, using ROI-aggregated responses and a manageable image subset for implementation.

Task: The project is structured in three main phases. In the first phase, we build a baseline encoding pipeline by extracting image features from at least two pretrained visual backbones (e.g., DINOv2 and CLIP) and training simple regressors to predict fMRI responses for selected ROIs. In the second phase, we compare three strategies: using all features, applying a generic pruning strategy, and applying the proposed ROI-aware and semantics-aware pruning strategy. The goal is to measure how much performance can be retained while reducing the number of features, and to identify a “maximally minimal” model for each ROI. In the third phase, we analyze how different ROIs and semantic image groups rely on different features and, if time permits, explore extensions such as subject-aware pruning, caption-guided grouping, or lightweight ROI-gated combinations of backbones.

Main objectives:

- *Baseline ROI encoding pipeline:* Build a reproducible pipeline that extracts pretrained visual features and predicts ROI-level fMRI responses. Report metrics such as Pearson correlation or R^2 on held-out data.
- *ROI-aware and semantics-aware pruning:* Compare full features, global pruning, and the proposed pruning strategy. Evaluate the trade-off between predictive performance, compactness and interpretability.
- *Backbone comparison and maximally minimal models:* Study whether different ROIs prefer different backbones or layers and estimate the minimum number of features needed to preserve strong encoding performance for each ROI.
- *Optional exploratory extensions:* Investigate subject-aware pruning, simple subject-adaptive ROI gating, ROI-gated ensembles of backbones, or retrieval-based / CLIP-based validation of the semantic coherence of selected features.

References:

- [Mahdiani, Morteza, and Ian Charest. "VISGate: ROI-Conditioned Dual-Head Encoders that Align Visual Features and Brain Responses." UniReps: 3rd Edition of the Workshop on Unifying Representations in Neural Models.](#)
- [Uzair Hussain and Kamil Uludag. "Boldreams: Dreaming with pruned in-silico fmri encoding models of the visual cortex." arXiv preprint arXiv:2501.14854, 2025C](#)